

Problem of the Week

Due Monday, September 28 at 11:59 pm (Arizona time)

A Product of Four

Find all real numbers x such that

$$(x + 1)(x + 3)(x + 5)(x + 7) = 9.$$

Prove that your list of solutions is complete.

How to enter

Scan the code, or go to ac.nau.edu/lms-apps/self-enroll/545026, to join the POTW Canvas course. Then upload **one PDF** (typed, or neatly handwritten and scanned) to this week's assignment. Open to all NAU undergraduates.

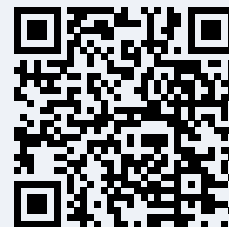
The rules

- **Your own work only.** No AI tools (ChatGPT, Gemini, Claude, Copilot, Photomath, ...), no internet, no calculators or software, and no help from other people.
- **Explain everything.** A bare answer earns little credit; a clear, complete argument earns full credit.
- Any participant may be asked to explain their solution in person.

Scored 0–4. Full rules, past problems, and model solutions:

naumathstat.github.io/problem-of-the-week

Questions: ian.williams@nau.edu



Join on Canvas